

Lab Manual — Operators in Python

Module 1 | Python Programming Fundamentals

Topic 3 of 3: Operators

Background

An **operator** is a symbol that performs an operation on one or more values (called operands). Python has several categories of operators:

Arithmetic Operators

Operator	Operation	Example	Result
+	Addition	5 + 3	8
-	Subtraction	5 - 3	2
*	Multiplication	5 * 3	15
/	Division (float)	5 / 2	2.5
//	Floor division	5 // 2	2
%	Modulus (remainder)	5 % 2	1
**	Exponentiation	2 ** 3	8

Comparison Operators

Return True or False. | Operator | Meaning | |-----|-----| | == | Equal to | | != | Not equal to | | > | Greater than | | < | Less than | | >= | Greater than or equal | | <= | Less than or equal |

Logical Operators

Operator	Meaning	Example
and	Both must be True	x > 0 and x < 10
or	At least one True	x < 0 or x > 100
not	Inverts the result	not True → False

Assignment Operators

Operator	Equivalent
x += 5	x = x + 5
x -= 5	x = x - 5
x *= 2	x = x * 2

Operator	Equivalent
<code>x //= 2</code>	<code>x = x // 2</code>
<code>x **= 2</code>	<code>x = x ** 2</code>

Solved Tasks

Task 1 — Arithmetic Operations

Objective: Use all arithmetic operators on two numbers.

```
a = 17
b = 5

print(f"{a} + {b} = {a + b}")
print(f"{a} - {b} = {a - b}")
print(f"{a} * {b} = {a * b}")
print(f"{a} / {b} = {a / b}")
print(f"{a} // {b} = {a // b}")
print(f"{a} % {b} = {a % b}")
print(f"{a} ** {b} = {a ** b}")
```

Output:

```
17 + 5 = 22
17 - 5 = 12
17 * 5 = 85
17 / 5 = 3.4
17 // 5 = 3
17 % 5 = 2
17 ** 5 = 1419857
```

Task 2 — Comparison Operators

Objective: Compare two values and observe boolean results.

```
x = 10
y = 20

print(f"x == y : {x == y}")
print(f"x != y : {x != y}")
print(f"x > y : {x > y}")
print(f"x < y : {x < y}")
print(f"x >= y : {x >= y}")
print(f"x <= y : {x <= y}")
```

Output:

```
x == y : False
x != y : True
x > y : False
x < y : True
x >= y : False
```

```
x <= y : True
```

Task 3 — Logical Operators

Objective: Combine conditions using and, or, not.

```
age = 20
has_id = True

# Both conditions must be True
can_enter = age >= 18 and has_id
print("Can enter club:", can_enter)

# At least one condition True
is_eligible = age < 15 or age > 60
print("Eligible for discount:", is_eligible)

# Negate a condition
is_minior = not (age >= 18)
print("Is minor:", is_minior)
```

Output:

```
Can enter club: True
Eligible for discount: False
Is minor: False
```

Task 4 — Assignment Operators

Objective: Modify a variable in-place using compound assignment operators.

```
score = 100
print("Start:", score)

score += 20
print("After += 20:", score)

score -= 10
print("After -= 10:", score)

score *= 2
print("After *= 2:", score)

score //= 3
print("After //= 3:", score)

score **= 2
print("After **= 2:", score)
```

Output:

```
Start: 100
After += 20: 120
After -= 10: 110
After *= 2: 220
After //= 3: 73
After **= 2: 5329
```

Task 5 — Practical: Even or Odd Checker

Objective: Use the modulus operator to determine if a number is even or odd.

```
numbers = [0, 1, 7, 14, 25, 100, -3]

for n in numbers:
    if n % 2 == 0:
        print(f"{n:4} → Even")
    else:
        print(f"{n:4} → Odd")
```

Output:

```
0 → Even
1 → Odd
7 → Odd
14 → Even
25 → Odd
100 → Even
-3 → Odd
```

Student Practice Tasks

Solve the following tasks on your own. No solutions are provided here — attempt each one independently.

1. Write a program that takes two numbers and prints the result of all seven arithmetic operations (+, -, *, /, //, %, **).
2. Calculate the area and perimeter of a rectangle using variables for length and width. Use only arithmetic operators.
3. Write a program that converts a temperature from Celsius to Fahrenheit. Formula: $F = (C * 9/5) + 32$. Test with 0, 100, and 37 degrees Celsius.
4. A student scored `marks = 73` out of 100. Use comparison operators to check if the student: passed (≥ 50), got a distinction (≥ 80), and failed (< 50). Print all three results.
5. Write a program that uses the modulus operator (%) to check if a number is divisible by both 3 and 5, only by 3, only by 5, or neither. Test with the numbers 15, 9, 10, and 7.
6. Given `x = 8` and `y = 3`, use logical operators to evaluate and print: `x > 5 and y > 5`, `x > 5 or y > 5`, and `not (x == y)`.
7. Write a bank balance simulator. Start with `balance = 5000`. Use += to deposit 1500, -= to withdraw 800, and *= to apply a 1.05 interest multiplier. Print the balance after each step.
8. Use the exponentiation operator (**) to compute and print: 2 to the power of 10, the square root of 144 (hint: `144 ** 0.5`), and 3 cubed.

9. Write a program that uses only comparison and logical operators (no `if` statements) to determine whether a given year is a leap year. A year is a leap year if it is divisible by 4 but not 100, OR divisible by 400. Print `True` or `False`.
10. Combine all operator types in one program: start with a number, apply arithmetic to transform it, compare it to a threshold, and use a logical operator to combine two comparison results into a final boolean output. Print each step.